

The Price of War

Jonathan Federle, André Meier, Gernot J. Müller,
Willi Mutschler, Moritz Schularick

Presented by Maciej Batkowski

AER, 2026

- **Context:** Rising geopolitical tensions and the return of interstate wars.
- **Question:** Who bears the economic price of war?
- **Traditional focus:** The economic destruction of the war site (e.g., Barro, 2006).
- **This paper's focus:** The *international spillovers* of war on non-war sites, analyzing how geography and trade integration dictate the economic fallout.

Key Contributions

- 1 **Novel Historical Dataset:** 150 years of data (1870–2022) covering 60 countries and 176 war sites.
- 2 **Empirical Estimates:** Quantifies the dynamic effects of war on output and inflation for the war site (Home) and foreign countries (Nearby vs. Distant).
- 3 **Structural Interpretation:** Develops an international business cycle model to explain *why* and *how* war shocks spill over to neighboring countries.

Data and Identification

- **Data Sources:** Correlates of War (COW) project, Jorda-Schularick-Taylor Macrohistory Database.
- **War Definition:** Baseline focus on “large war sites” (casualties $> 10,000$).
- **Identification Strategy:** Wars are typically driven by nationalism, ideology, or long-term geopolitical goals (exogenous to the short-term business cycle).
- **Frequent & Exogenous:** Exposure to neighboring wars occurs almost at business cycle frequency, and narrative analysis confirms their onset acts as an exogenous shock independent of economic conditions.

Table 1: Categories of war sites

Severity	Casualties		Length		Wars	Macro time-series for...	
	Min.	Mean	Mean	Median	Total	Home	Foreign
All sites	46	184,357	2.6	2.0	176	66	2,786
Large sites	10,000	334,920	3.3	2.0	96	38	1,798
Major sites	105,525	658,323	4.4	4.0	46	21	1,026

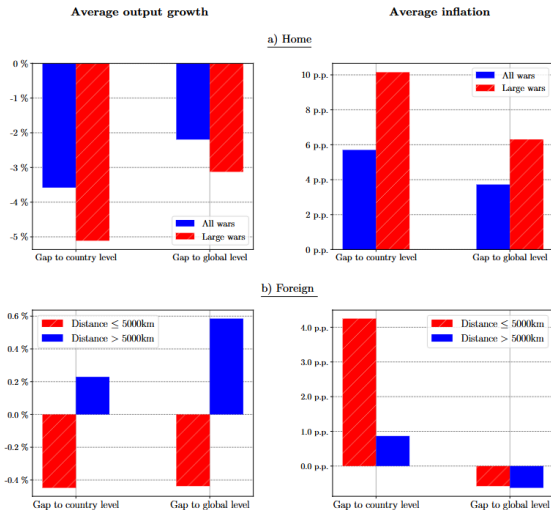
Notes: Table shows different war-site samples according to casualty thresholds. Minimum casualties denotes war site with lowest number of casualties in sample. Length denotes duration in years for wars in our sample. Home (Foreign) are observations for when a war starts in the war site (abroad), provided macro time series are available. Large sites

Macroeconomic Performance After War Onset

- Benchmarks 5-year average growth and inflation against historical country norms and global norms.
- **Home (War Site):** Severe output shortfalls (up to 5 p.p. below norm) and excess inflation (up to 10 p.p. above norm).
- **Foreign (Spillovers):** Adverse effects (lower growth, higher inflation) are heavily concentrated in nearby economies ($\leq 5000\text{km}$).

Macroeconomic performance

Figure 3: Macroeconomic performance after start of wars



- Local projections (Jordà, 2005) to estimate macroeconomic effects over an 8-year horizon.
- **Baseline Specification:**

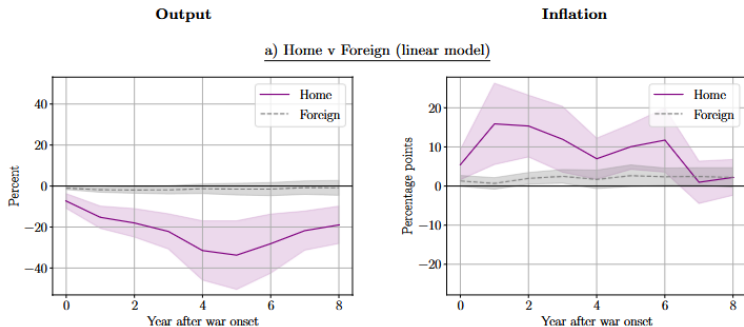
$$x_{i,t+h} - x_{i,t-1} = \alpha_{i,h} + \gamma_h \text{Home}_{i,t} + \psi_{D,h} F(i, t) \text{Foreign}_{i,t} + \psi_{N,h} [1 - F(i, t)] \text{Foreign}_{i,t} + u_{i,t+h}$$

- Smooth-transition framework assigns foreign countries to “Nearby” (N) or “Distant” (D) regimes based on geographic or trade distance $F(i, t)$.

$$0 \leq F(i, t) = \frac{\ln(1 + d_{i,t})}{\ln(1 + d_{\max})} \leq 1.$$

The Macroeconomic Impact: The War Site (Home)

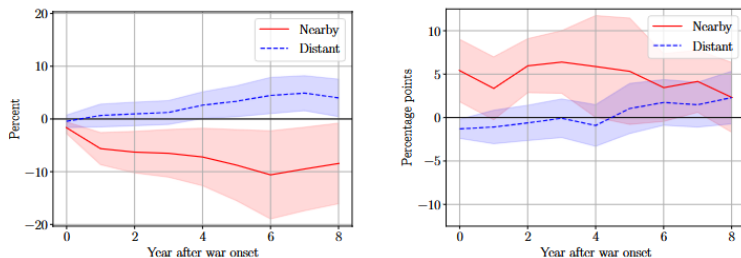
- War acts as a massive adverse supply shock on the war site.
- **Output:** GDP falls by roughly 30% relative to trend over five years.
- **Inflation:** Surges by 15 percentage points.



International Spillovers: Nearby vs. Distant

- Spillovers are highly dependent on geographic proximity.
- **Nearby Countries:** Suffer a 10% output drop and a 5 percentage point increase in inflation.
- **Distant Countries:** Experience mild positive output spillovers (winners and losers of war).

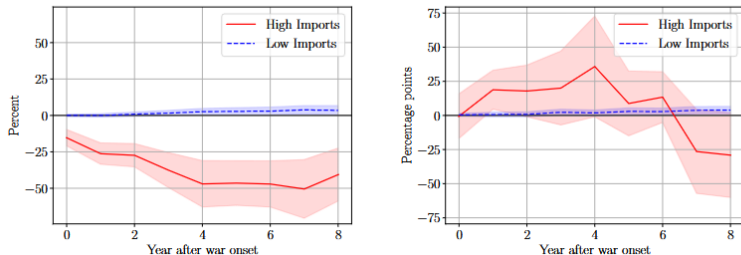
b) Spillovers to Foreign: Nearby v Distant (smooth transition model)



The Role of Trade Integration

- Replacing geographic distance with **trade distance** yields even starker results.
- Countries with high pre-war imports from the war site experience massive economic contractions.
- Suggests that the disruption of trade and supply chains is the primary transmission mechanism.

b) Accounting for trade exposure to war site



Robustness and Other Mechanisms

- **Belligerents vs. Third Countries:** The adverse spillovers affect nearby third-party countries almost as severely as belligerents.
- **Military Spending:** Increases globally, but much more in the war site.
- **Capital and TFP:** Capital stock drops heavily in Home (20%) and Nearby (5%). TFP collapses in Home.

Variable	Home (War Site)	Nearby	Distant
Output Drop	-30%	-10%	+2%
Inflation Spike	+15 pp	+5 pp	~ 0 pp
Capital Stock	-20%	-5%	0%

The Structural Model I: Households and Capital

- **Preferences:** Households maximize expected lifetime utility over consumption and labor, featuring habit formation and sticky wages

$$U_{j,t} = E_0 \sum_{t=0}^{\infty} \beta^t \left(\log(C_{j,t} - h\bar{C}_{j,t-1}) - \frac{\kappa_L}{1 + \sigma_L} L_{j,t}^{1+\sigma_L} \right)$$

- **The War Shock Process:** The war shock Ξ_t is an AR(2) process triggered by an innovation η_t at the onset of the war

$$\Xi_t = \rho_1 \Xi_{t-1} + \rho_2 \Xi_{t-2} + \eta_t$$

- **Capital Destruction:** The war shock Ξ_t exogenously destroys a fraction of the physical capital stock in the war site

$$K_{j,t} = \left((1 - \delta_K) K_{j,t-1} + \left[1 - \frac{\phi_K}{2} \left(\frac{l_{j,t}}{l_{j,t-1}} - 1 \right)^2 \right] l_{j,t} \right) e^{-\Delta_j^K \Xi_t}$$

The Structural Model II: Production and Trade

- **Production & Trade:** Firms use capital, labor, and intermediate inputs $X_{j,t}$ (which embeds the trade integration channel) to produce varieties under monopolistic competition

$$A_{j,t}(X_{j,t}(m))^{\alpha_X} [K_{j,t-1}(m)^{\alpha_K} L_{j,t}(m)^{1-\alpha_K}]^{1-\alpha_X} = \sum_{i \in \{H,N,D,R\}} Y_{ji,t}(m)$$

- **Productivity Decline:** Total Factor Productivity (TFP) is persistently reduced by the same war shock in the Home economy

$$\log(A_{j,t}/A_j) = \rho_A \log(A_{j,t-1}/A_j) - \Delta_j^A \Xi_t$$

- **Final Goods Assembly:** Wholesale goods are aggregated with a constant elasticity of substitution σ , subject to import adjustment costs $\Gamma_{ij,t}$

$$Y_{j,t} = \left[\sum_{i \in \{H,N,D,R\}} \gamma_{ij}^{\frac{1}{\sigma}} [\Gamma_{ij,t} Y_{ij,t}]^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

The Structural Model III: Policy and Market Clearing

- **Military Spending:** Government spending is funded via lump-sum taxes and surges globally in response to the war shock innovation η_t

$$\log(G_{j,t}/G) = \rho_G^j \log(G_{j,t-1}/G) + \Delta_j^G \eta_t$$

- **Monetary Policy:** The central bank adjusts interest rates following a Taylor rule targeting inflation $\Pi_{j,t}$ and the GDP gap

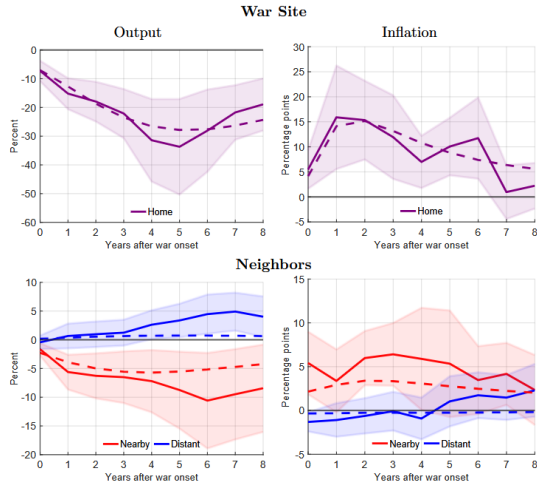
$$\frac{R_{j,t}}{R_j} = \left(\frac{R_{j,t-1}}{R_j} \right)^{\rho_R^j} \left[\left(\frac{\Pi_{j,t}}{\bar{\Pi}_j} \right)^{\psi_\Pi^j} \left(\frac{\text{gdp}_{j,t}}{\text{gdp}_j} \right)^{\psi_{\text{gdp}}^j} \right]^{1-\rho_R^j}$$

- **Market Clearing:** Final goods are absorbed by consumption, investment, intermediate inputs, government spending, and adjustment or bond holding costs

$$Y_{j,t} = C_{j,t} + I_{j,t} + X_{j,t} + G_{j,t} + a[u_{j,t}]K_{j,t-1} + \frac{\phi_B}{2} \left(\frac{B_{R,j,t} ER_{j,t}}{P_{j,t}} \right)^2$$

Model Validation: Empirical vs. Theoretical Responses

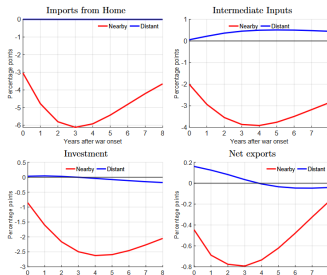
Figure 9: The macroeconomic impact of war: model v data (targeted)



Inspecting the Transmission Mechanism

- Why do nearby economies suffer despite no physical destruction?
- **The Trade Channel:**
 - 1 Home's supply side contracts (capital/TFP destroyed).
 - 2 Price of Home exports surges.
 - 3 Nearby economies slash imports from Home.
 - 4 **Scarcity of intermediate inputs** forces a contraction in Nearby's production and an endogenous drop in investment/capital.

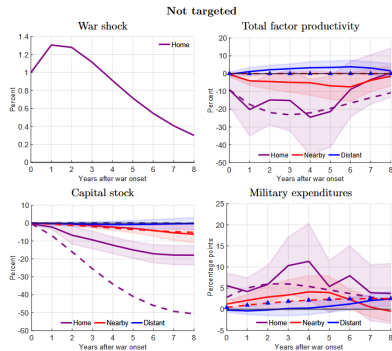
Figure 11: The transmission of the war shock



Model Validation

- The model accurately reproduces the un-targeted dynamics of the capital stock and TFP.
- Demonstrates that the endogenous reduction of capital in Nearby is a rational response to the war shock in Home.

Figure 10: The macroeconomic impact of war—model validation



- **Asymmetric Global Spillovers:** The economic burden of war is dictated by proximity. War sites suffer a $\sim 30\%$ output collapse, and nearby countries face a severe 10% decline, while distant economies experience mild positive spillovers.
- **Key Transmission Mechanisms:** The adverse spillovers to neighboring economies are primarily driven by the disruption of integrated trade networks (specifically intermediate inputs) and endogenous global surges in military spending.
- **Persistent Stagflationary Shock:** War acts as a devastating supply-side shock characterized by physical capital destruction and prolonged TFP declines, driving up inflation while persistently depressing output.
- **Policy Implications:** Central banks and governments in neighboring countries cannot treat nearby wars as transitory shocks; they face a prolonged stagflationary environment that requires managing disrupted supply chains alongside increased fiscal burdens.